

TAYLOR'S UWE DUAL AWARDS PROGRAMMES SEPTEMBER 2026 SEMESTER

ITS69304 - Data Analytics and Machine Learning Group Assignment (30%)

DUE DATE: via MyTIMeS

Instruction to Candidates:

1. Name your answer file as ITS69304_XXXXXX_GROUPASGNMT.pdf where XXXXXX is your GROUP NUMBER. Then, submit to the MyTIMeS portal via the link "GROUP ASSIGNMENT submission" on the module page. (Do not submit the question paper).

STUDENT DECLARATION

1. I confirm that I am aware of the University's Regulation Governing Cheating in a University Test and Assignment and of the guidance issued by the School of Computing and IT concerning plagiarism and proper academic practice and that the assessed work now submitted is in accordance with this regulation and guidance.
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Section	Marks
Answer all questions.	30 Marks
Learning Outcomes The objective of this assessment is to enable the students to: 1. MLO2: Analyse different data analytics techniques for a given application	

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Digitalisation Issues in TNB: Data Analytics & AI Solutions

Proposal (Theme D: Customer Digital Experience)

1. Introduction & Background

1.1 Problem Identification and Criticality

Tenaga Nasional Berhad (TNB) has transitioned toward a digital-first strategy via the myTNB ecosystem. However, the success of this digital transformation is currently hindered by a reactive customer satisfaction framework. Presently, TNB relies heavily on voluntary ratings and manager-led feedback, which creates a silent majority of dissatisfied users who abandon the digital platform without providing explicit feedback. The main challenge that the company is facing is the lack of an automated and intelligent way of analysing the feedback and behaviour of the customers. Without the use of an AI-based method of analysing the feedback and behaviour of the customers, the company is experiencing an invisibility of the customer experience. This invisibility of the customer experience is an important challenge for the company in the following areas:

- **Cost Efficiency:** The company's poor digital experience forces the customers to use offline channels. This means that the company is spending on maintaining offline channels while spending on ineffective digital channel updates. Additionally, the company is spending on acquiring new customers since the customers' churn is unpredictable.
- **Operational Reliability:** Currently, the company uses either manual or inflexible systems to direct the customers' complaints. This means that if an important technical issue exists among the billing questions, the company will be forced to take longer to address the concern.
- **Customer Trust and Corporate Image:** In this digital era, the company trust is built on Responsiveness and Assurance (Azmi, 2023). This means that the company's invisibility of the customer experience hurts the company's corporate image. The company's customers will feel that their feedback and complaints have been sent to the digital black hole. This will lead to the customers' mistrust of the company's digital capabilities.

Table 1 Impacts of Problem

Impact Area	What happens if problem is not solved	What improves if solved
Cost	High operational overhead; redundant offline service costs.	Reduced manual sorting; efficient, data-backed app updates.
Reliability	SLA breaches due to manual routing delays, which leads to silent churn.	85-95% routing accuracy with proactive churn identification.
Safety	High-risk problems are hidden in the complaint queues.	Real-time early warnings and emergency prioritization to prevent escalation of danger.

Customer trust	Corporate image and customer relationship negatively impacted	Instant complaint routing, personalized retention and faster resolution.
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1.2 Proposed Solution (High-Level Overview)

This proposal recommends an Integrated AI-Driven Customer Retention and Operational Efficiency System. This system utilizes machine learning models in transforming customer data and corresponding complaints into useful insights, particularly tailored for TNB's digital transformation.

- **Automated Complaint Categorization & Routing:** This system utilizes a Random Forest model with TF-IDF Vectorization, enabling the system to automatically sort customer complaints based on the six major departments in the company. It replaces the traditional manual sorting process, allowing for reduced errors and proper routing of complaints to the right technical teams for a 24-hour response cycle.
- **Proactive Churn Mitigation & Decision Support:** This system utilizes an XGBoost Classifier model on customer data, enabling the system to identify and classify potential customer churn. It acts as a decision support system for management, particularly in updating corporate strategies based on data-driven information on customer churn, such as high levels of service interruptions and low levels of payment reliability.
- **Quality Measurement:** This system measures Service Quality (SQ) and Trust (T), particularly in line with industry standards (Nurul Amira Azmi, 2023). It measures Reliability (tracking the number of service interruptions) and Responsiveness (tracking automated resolution times).

By merging automated NLP (Natural Language Processing) for complaints with predictive analytics for churn, this solution enhances the Technology Support (TS) and Corporate Image (CI) of TNB, moving from a reactive to a predictive customer service model.

1.3 Comparison with Existing Industry and Academic Solutions

Solution 1: IBM watsonx Assistant (Industry)

IBM watsonx Assistant is an enterprise AI platform widely deployed for automated customer complaint management, offering NLP-based intent classification, sentiment analysis, and customer service routing. However, Watsonx Assistant's LLMs are limited in their applicability in multilingual business environments as they are only available in English (IBM, 2024). This makes it unfavourable for the Malaysian context, as it offers no native handling of code-switching between Malay and English, which is common in customer feedback. Furthermore, its intent-based architecture is designed for query resolution rather than multi-class complaint classification, urgency scoring, or resolution time prediction.

Solution 2: Sentiment Identification for Malay-English Code-Switched Data (Academic)

Mohd Shamsuddin et al. (2024) developed a semi-automatic sentiment identification framework for Malay-English code-switched social media data, demonstrating the complexity of multilingual sentiment distribution in Malaysian digital communication. While valuable for addressing code-switching challenges, the study is limited to social media text and does not extend to complaint classification, urgency scoring, or operational routing.

Proposed Improvement

Our approach advances beyond simple text analysis and becomes a full-fledged business intelligence solution. The architecture leverages context-aware NLP that supports Malay-English code switching to attain 85-95% routing accuracy for the teams specific to TNB. Instead of using pre-built tools and solutions, the architecture uses sentiment as input to an XGBoost-based churn predictor. By using Random Forest for routing and predictive behavioral analytics, the architecture demonstrates the link between service quality and loyalty and provides a tailored and multi-lingual solution that outperforms industrial solutions and academic research above.

1.4 Success KPIs

Table 2 Business and Technical KPIs

KPI	Target
Average complaint resolution time	$\leq 30\%$
Complaint misrouting rate	$\leq 40\%$
MyTNB app user retention rate	$\geq 20\%$
Offline/walk-in service requests	$\geq 25\%$
Customer Satisfaction Index (CSI)	$\geq 15\%$
Complaint classification (accuracy, prediction, recall, f1-score)	$\geq 88\%$
Churn prediction (accuracy, prediction, recall, f1-score)	$\geq 88\%$

2. Data Modelling, ML Algorithms & Requirements

2.1 Proposed Dataset and Data Model

Table 3 Key Data Sources

Data Source	Description	Frequency	Volume
TNB Complaints Data	Structured data regarding complaints submitted by customers	Daily	2000 records (6 months)
TNB Customer Churn	Structured behavioral and usage data for retention analysis	Daily	2000 records (6 months)

Data Quality Issues:

- Missing complaint text (incomplete submissions)
- Mixed-language complaint data (Malay/English)
- Duplicate submissions (same customer complaint through multiple channels)

Table 4 Data Dictionary of TNB Complaints Data

Field	Type	Example	Notes
complaint_text	string	"Late payment fee is too high."	Customer's complaint content
category_actual	string	"Billing"	Complaint's category
resolution_time_actual	double	24.0	Complaint's resolution time (hour)
urgency_score	int	51	Complaint's urgency [0(not urgent) - 100(very urgent)]

Table 5 Data Dictionary of TNB Customers Churn Data

Field	Type	Example	Notes
account_type	string	"residential"	Customer's account type
monthly_consumption_kwh	int	600	Customer's monthly electricity consumption in kilowatt-hour
complaint_count_6m	int	2	Customer's complaint counts for the last six months
churn	boolean	0	Customer churn (0: no, 1: churn)

2.2 Data Model (Entities & Relationships)

The system follows a relational structure where the *customer_id* acts as the main connection across the datasets

Entities:

1. Complaint
2. Customer
3. Churn Status

Relationships:

- Customer (1) → Complaint (N): A single customer can submit multiple complaints over time.
- Customer (1) → Churn Status (1): Each customer is identified as churn or not.

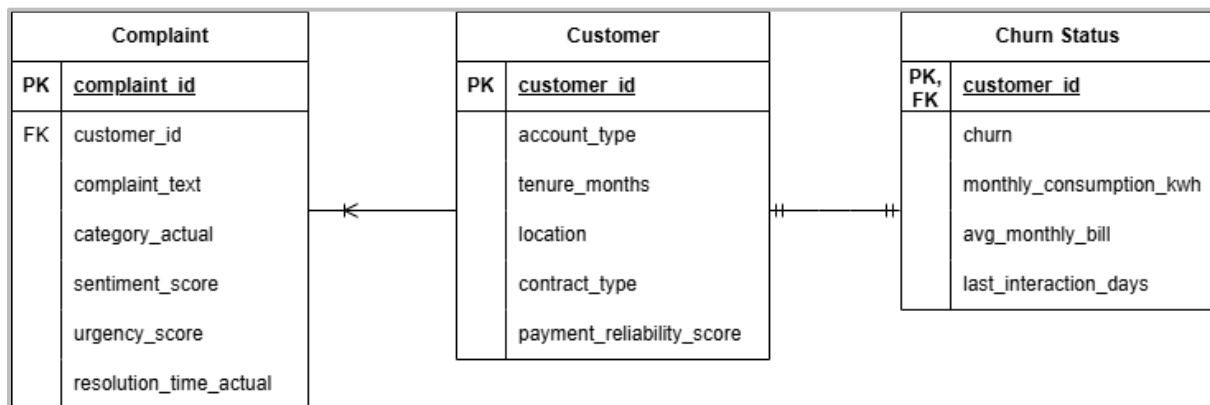


Figure 1 Entity Relationship Diagram

2.3 Machine Learning Algorithms

Model 1: Random Forest Classifier (Complaint Classification)

- Perform well when working with big and complex data with many features
- High accuracy while prevent overfitting
- Ability to analyse feature importance

Random Forests were used to categorize complaints into 6 classes (Billing, Service...). The model performed well here because they were able to handle the high-dimensional sparse data generated by TF-IDF vectorization without being prone to overfitting compared to a single decision tree.

Model 2: XGBoost Classifier (Churn Prediction)

- Achieve high accuracy by capturing complex interactions between features
- Good in handling missing values and outliers
- Scalable for large datasets

XGBoost algorithm uses 14 different customer features to predict customer churn. Its advanced gradient boosting framework helps capture the non-linear relationship between consumption patterns and churn intention.

2.4 Machine Learning Pipeline

1. Data Preprocessing

- Direct remove or mean imputation for missing value
- Duplicate removal

2. Feature Engineering

- Text cleaning: lowercasing and special character removal from *complaint_text*
- Extract text (*text_length*, *word_count*), temporal (*hour*, *is_weekend*), derived (*bill_per_kwh*, *high_complaint_flag*) features
- Encoded categorical features

3. Label Definition

- *category_actual* (Billing, Service...) as the label for complaints data
- *churn* (0, 1) as the label for churn data

4. Model Training

- Train-test split: 80/20
- TF-IDF features for complaint text
- Train the model
- Make predictions on test set

5. Evaluation

- Classification report: accuracy, precision, recall, f1-score,
- Confusion matrix

6. Deployment

- Auto classification via API calls
- Monthly retraining with production data

2.5 Evaluation Metrics

Table 6 Classification Metrics

Metrics	Interpretation
Accuracy	High-level and straightforward overview of model correctness across all categories
Precision	• Minimize sending complaint to wrong department (waste resolution time) • Ensure TNB does not waste expensive retention offers
Recall	• Minimize misclassification of categories (bad resolution) • Minimize the revenue loss caused by customers who leave unnoticed.
F1-score	Balance performance of all categories
Confusion Matrix	Visualizes the overall true & false positives, and true & false negatives

3. Data Processing Technologies

By transitioning to a predictive customer service model, a hybrid architecture is designed to support the high volume of historical data (model training) and incoming requests from customers (real-time routing and alerts).

3.1 Data Acquisition and Processing Pipeline

Real-Time Data Processing (Fast Layer):

Stream ingestion via API Gateway (myTNB app complaints) and IoT Gateway (smart meter consumption data). Real-time scoring of the pre-trained Random Forest model and the XGBoost model is monitored and triggered immediate routing of urgent "Safety" or "Outage" complaints or high-risk churn notifications.

Batch Data Processing (Batch Layer):

Nightly ingestion of structured data from TNB's historical databases (payment history, tenure, resolution logs). ELT is then performed, where raw data is cleaned, mixed language pre-processing occurs, and features are engineered to retrain models. Schedule weekly retraining of models to adapt to changing customer behaviour.

3.2 Recommended Technology Stack

Table 7 Recommended Technology Stack

Layers	Technology	Justification
Data Ingestion	Apache Kafka	Kafka acts as a high-throughput, fault-tolerant "buffer" that decouples data sources from sensors (<i>Apache Software Foundation, 2026</i>).
Data Storage	AWS S3 (Data Lake)	Offers high durability and scalability for both raw logs and processed datasets at a low cost (<i>Amazon Web Services, 2026</i>).
Data Processing	Python, Pandas, NumPy	The industry standard for data manipulation; its high-level data structures allow for rapid cleaning and transformation of tabular data.
ML	Random Forest, XGBoost Classifier	Random Forest for category classification, while XGBoost for churn prediction.
Governance	Apache Ranger	Support multiple authorization methods and provide audit trails for compliance.
Model Serving	FastAPI	Serves the models as REST APIs for instant integration with the myTNB app.
Dashboard & Reporting	Power BI	Lower total cost of ownership (TCO) for organizations already utilizing the Microsoft ecosystem and its accessibility for non-technical business users. (<i>Henderson, 2025</i>).

Deployment Environment	AWS	Extensive global infrastructure, superior scalability, and mature machine learning ecosystem (<i>Global Infrastructure</i> , n.d.).
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3.3 System Architecture

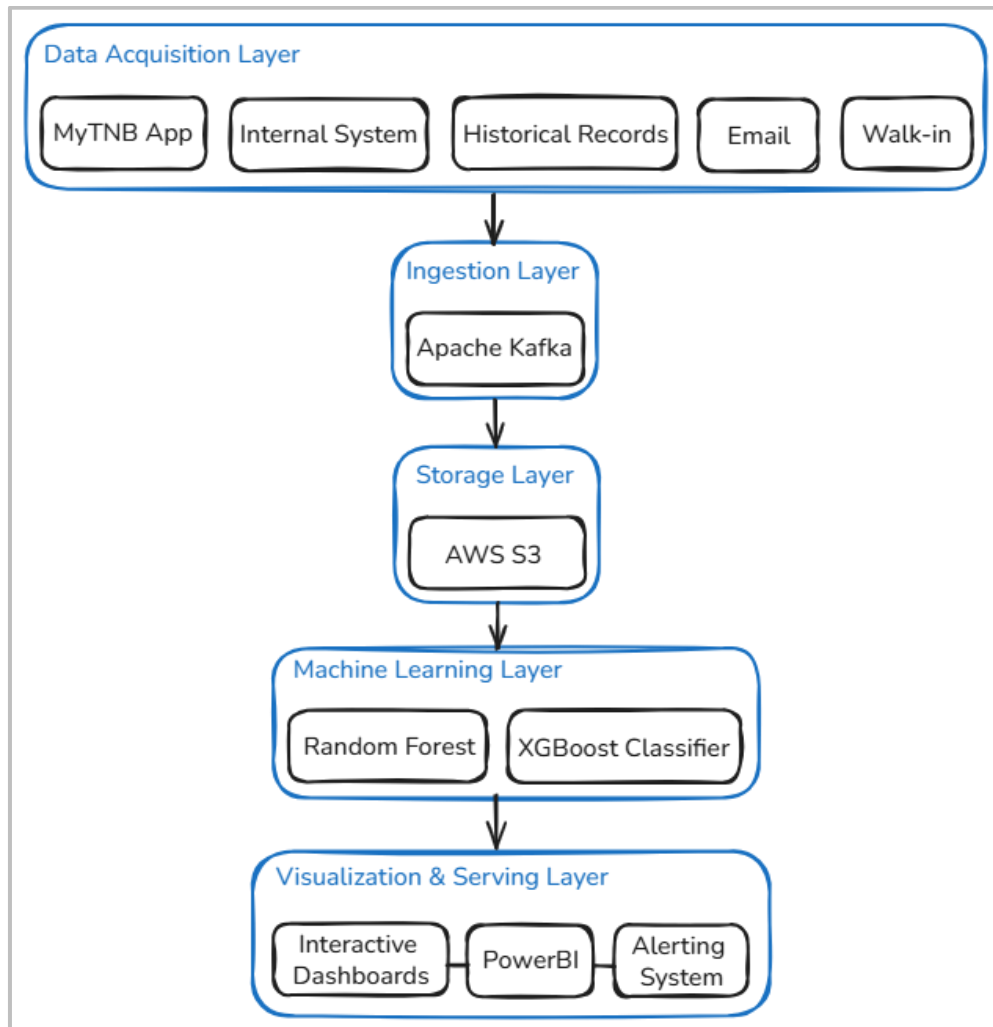


Figure 2 System Architecture Diagram

4. Reporting & Visualisation

To support data-driven decisions and live oversight of operations management, this research proposes an Intelligent Complaint Management System (iCMS), which incorporates multiple interactive data dashboards. These data dashboards use data visualization techniques to present complex data in an easily understandable format, helping management understand the overall system, trends in complaint data, and customer service activities in a simple manner.

These data dashboards are primarily intended to support Customer Service Supervisors and other management personnel to identify trends and improve resource management and efficiency in complaint handling through various activities. Using live data feeds and multi-dimensional data analysis capabilities, management can better monitor operations and make timely decisions.

4.1 Executive Dashboard Design

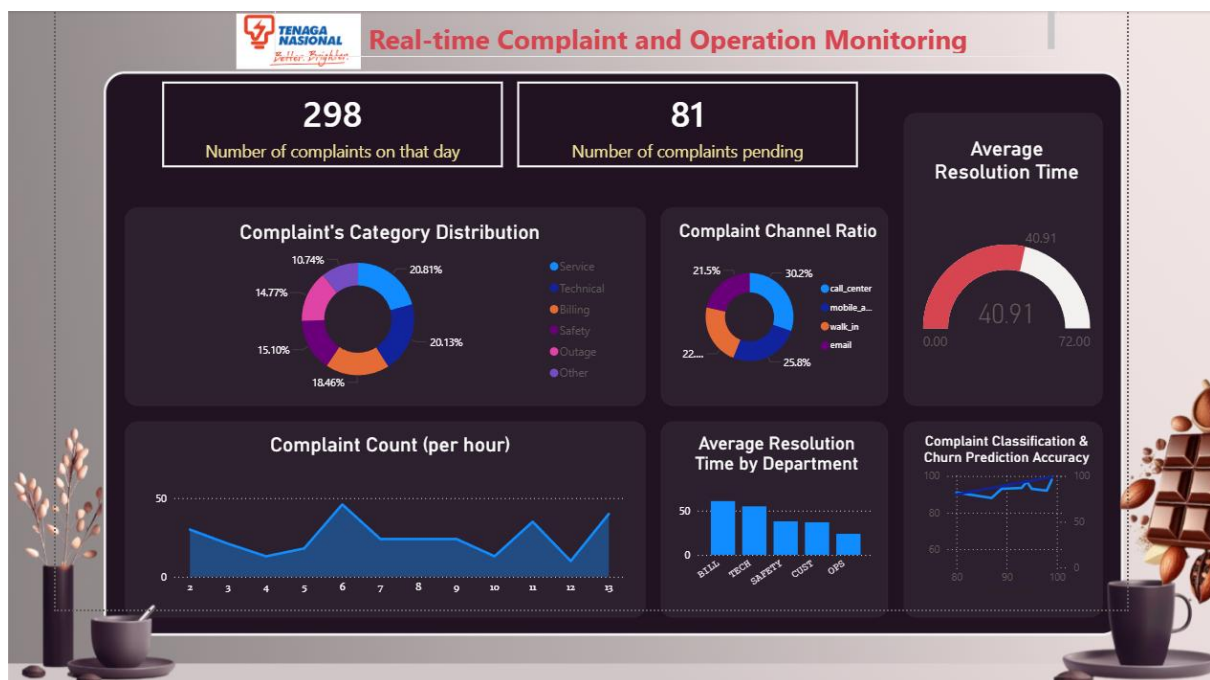


Figure 3 TNB Real-Time Complaint and Operation Monitoring Dashboard

The figure above indicates our executive dashboard that showcase multiple important operational details. It provides a convenient way to monitor complaint handling in real time and visualizes the complaint management process. Through various data visualization components, such as KPI cards, pie charts, bar charts, and trend graphs, the system can

comprehensively display actionable insights. Below is a brief overview of what you can expect to see:

- **Number of Complaints on that Day:** The dashboard will show you the total number of complaints that have been received by the system that day. It helps in determine the current customer feedback pressure and if there is an abnormal increase in the number of complaints
- **Number of Complaints Pending:** The dashboard will show you the total number of complaints that are still pending. If this starts to climb, it could mean that the complaint resolution flow is getting overwhelmed or the system is slowing down.
- **Complaint's Category Distribution:** The dashboard will show you a donut chart that represents the different types of complaints that are being received by your system. These could range from Service to Technical to Billing to Safety to Outage to Other.
- **Average Resolution Time by Department:** A bar chart is used for comparing the average processing times for each of the departments, i.e., BILL, TECH, SAFETY, CUST, and OPS. This metric reflects the efficiency of each department to resolve complaints within the specified time.

4.2 Recommended Tools

In this task, processing timeliness is crucial to the system's success. **Power BI** connects directly into Azure SQL databases via Direct Query, so there's no need to go through an ETL process. And if we add an auto-refresh every 15 minutes and streaming data, TNB managers can see the queue and processing in real time on the Power BI dashboards. This almost instantaneous data availability, combined with Power Automate's notification features, allows TNB managers to respond in an instant when complaint volume surges, moving from post-event analysis to real-time control.

4.3 Alert Thresholds

Table 8 Operational Thresholds and Corresponding Alerts

Threshold	Alert
Churn probability > 0.85	Automatically flag the account in the CRM and allow manual intervention

Category == 'Safety' AND urgency score > 0.9	Bypass the standard queue and send SMS/push notifications directly to the Emergency Response Team (OPS).
Resolution time > SLA limit	Alert head of the department and suggesting a temporary reallocation of staff

5. Conclusion

Overall, the proposed AI-based system demonstrates a strategic transformation for TNB from a reactive "rating-based" feedback model to a proactive predictive analytics model. The integration ML models, such as Random Forest for automated complaint routing and XGBoost for proactive churn enable TNB to overcome the digital visibility gap while ensuring that critical safety and service issues are addressed, while customers are retained. Not only than that, this system optimizes operational costs by eliminating the need for physical channels but also help in building greater customer trust by resolving issues accurately and in a timely manner. Finally, we believe this framework provides a scalable and reliable data-driven foundation for TNB's digital transformation and service excellence.

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